Research basis

Four studies inform the proposed retrieval and evidence checks

BGE-M3 motivates a multilingual retrieval comparison. Self-RAG motivates explicit evidence-support checks, although our pipeline does not reproduce its training procedure.

ALCE separates citation support from a citation simply existing. Lost in the Middle motivates a development test of evidence position.

Evidence behind the proposed method

Four studies published in 2023 and 2024

Study	Research concern	TenderGuard response
BGE-M3 Chen et al., 2024	Multilingual evidence retrieval	Compare lexical search, BGE-M3 and E5 on common chunks
Self-RAG Asai et al., 2024	Whether evidence supports a generated claim	Add support checks and measure false passes
ALCE Gao et al., 2023	Citation support and completeness	Check both source locations and support for each finding
Lost in the Middle Liu et al., 2024	Sensitivity to evidence position	Test position on development data, then freeze ordering

Research takeaway:

RU/KK tender performance requires our own evaluation

docs/literature-review/literature-review.md

Project implications

1. Compare retrieval on common chunks
2. Measure evidence support on our own corpus

Research questions

Tender requirements and company evidence

The main ML questions concern finding evidence and assessing the company against it. RQ1 compares the retrieval configurations. RQ2 compares one generation call with explicit extraction, assessment and reporting stages.

RQ3 covers the project privacy boundary and requires work from Ramazan and Rassul as well as ML tests. RQ4 checks whether users understand the report and its evidence.

Research questions and observations

The planned evaluation covers four questions

Question	What we ask	What we observe
RQ1	Which retrieval method finds the evidence needed for assessment?	Evidence recall and complete evidence sets
RQ2	Does staged analysis improve supported company-specific findings?	Status quality, citations and resource use
RQ3	Do controls enforce organization and processing boundaries?	Blocked violations and useful authorized processing
RQ4	Can users inspect findings and resolve missing information?	Task completion and correct interpretation

Research takeaway:

Documentary fit supports a human decision and does not establish legal eligibility.

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Project implications

1. Report language slices separately
2. Exercise privacy and user interpretation alongside ML quality

Annotated example: warranty evidence

SYN-03 Synthetic reference example

Tender: full hardware warranty of at least 24 months from acceptance

The package does not establish which offer applies.
Keep both citations and request confirmation.

Competing company commitments

Full hardware warranty from acceptance

Company source	Quoted commitment	If confirmed applicable
SYN-C03-A-v1	“exactly 12 months from acceptance”	unmet
SYN-C03-B-v1	“exactly 36 months from acceptance”	met

Expected: unknown

Evidence interpretation:

Missing evidence also gives unknown. A policy block remains a processing outcome.

docs/dataset/dataset-and-annotation.md

Project implications

1. Retain both company citations
2. Request confirmation of the applicable offer

Baseline comparison

B1 and B2 share the generator snapshot, original evidence and validation rules

B2 adds calls. We count all tokens, failures and cost.

B0 names the PostgreSQL lexical retrieval baseline already included in RQ1. It returns passages, so status macro-F1 does not apply to B0.

The primary analysis experiment compares B1 and B2. We freeze the original evidence pack, its ordering and hash.

B0, B1 and B2

Retrieval reference and two analysis workflows

Variant	Method	Output	Comparison role
B0	PostgreSQL lexical retrieval	Ranked source passages	Retrieval reference
B1	Shared retrieval, then one generation call	Requirements, statuses and citations	Primary comparator
B2	Same original evidence, then typed extraction, assessment and reporting	Same final schema with intermediate records	Primary candidate

Comparison takeaway:

Fairness means equivalent original evidence and rules, not identical compute.

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Project implications

1. Keep adaptive retrieval separate
2. Retain failures and the complete model overhead

Evaluation measures and conditions

Provisional planning targets

Also report false passes, processing failures, latency and total model cost

Separate document families and RU/KK results. Freeze settings before final testing.

The collection plan targets 30 independent tender families, split into 18 development and 12 test families, with a six-family development pilot. These counts are planned.

Measures and provisional targets

Planning values require development validation

Measure	Target	Scoring unit
Complete-evidence success@5	At least 0.85	Answerable queries with a complete acceptable evidence set
Status macro-F1	At least 0.80	Equal-weight F1 across met, unmet and unknown on reference requirements
Supported gold coverage	At least 0.80	All gold obligations, crediting correct findings with adequate support

Measurement takeaway:

Supported gold coverage uses every reference obligation, so missing requirements cannot improve the score.

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Project implications

1. Keep related documents in one partition
2. Freeze configurations before held-out testing

Limitations and optional fine-tuning

Small samples limit generalization. Documentary fit does not establish legal eligibility.
Defined privacy tests support only the boundaries they exercise.

Fine-tuning is optional because the core question can be tested with pretrained models, retrieval and controlled prompts. We first need a permitted corpus and an error analysis showing what adaptation should improve.

Training would also require independent held-out data, hardware and time. A model can memorize published synthetic examples, so they cannot establish unseen-test improvement.

Core work and optional adaptation

Scope follows the agreed proposal and protocol

Core evaluation	Fine-tuning after the baseline
Pretrained models with evidence retrieval and validated outputs	A permitted training set and a measured error pattern must justify adaptation
Separate RU/KK results and document-family partitions	Compare an adapted model with the unchanged base model
Citations and explicit unknown outcomes	Check hardware, time and budget before committing

Scope takeaway:

A negative result for staging remains valid if the comparison is fair and the evidence is retained.

docs/technology-comparison/ml-technologies.md

Project implications

1. Local generation and OCR remain extensions
2. Final model measurements follow implementation